

Volatility Forecasting with Residual Learning

Eddie Liu | GARCH, LSTM, and empirical model evaluation

An additive LSTM reduces observed forecast MSE by **5.48%** versus GARCH with a fixed floor. But the uncertainty interval includes zero, and invalid raw variance forecasts remain a material limitation.

3,738

common test dates

178

walk-forward folds

2010-2024

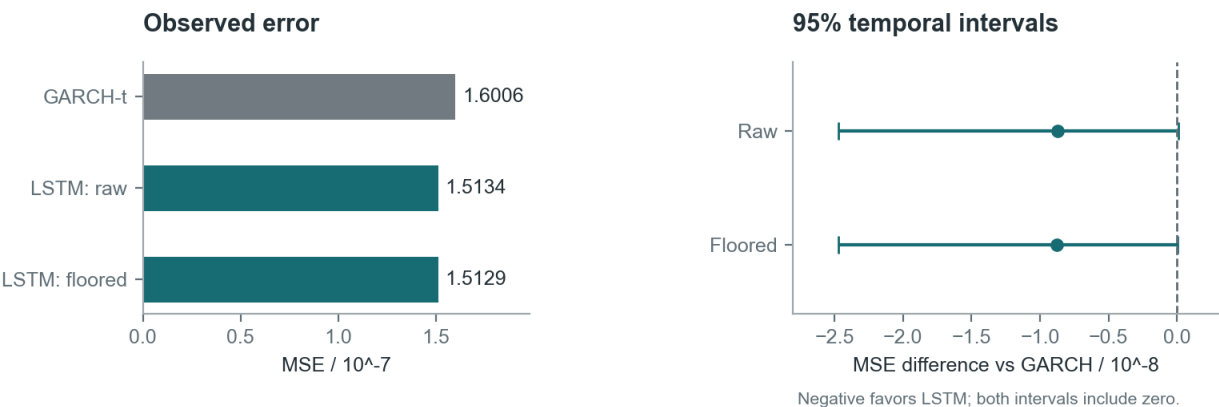
historical evaluation period

Can an LSTM learn GARCH's forecast errors?

The target is one-day S&P 500 variance, measured with squared daily log returns.

01 Forecast the baseline	02 Learn its errors	03 Evaluate both outputs
GARCH-t estimates next-day variance g using past returns.	LSTM predicts the signed error $e = y - g$ from lagged features.	Raw $h = g + \text{predicted } e$. Floored $h = \max(\text{raw } h, 1e-12)$.

The evidence at a glance



Paired stationary bootstrap: mean block length 21; 2,000 resamples. Intervals condition on the trained forecasts, not training-seed or model-selection uncertainty.

Raw MSE improves 5.45%, versus 5.48% when floored: the gain is not primarily a clipping effect.

Validity: 60 raw forecasts are nonpositive. Floored QLIKE is 457,801, versus -8.4613 for GARCH; lower is better. Lower MSE alone is not a deployable-model result.

Dynamic Portfolio Weighting

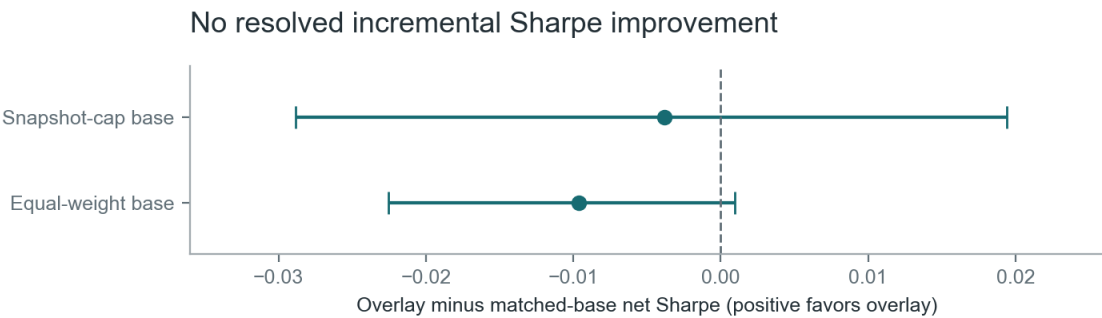
The legacy stock model applies an LSTM ratio correction to GARCH volatility. It is a separate model and sample from Part A, not a trading application of its additive-residual predictions.

The allocation rule

Weight = base weight x clip(trailing vol / saved forecast, 0.5, 1.5), then normalize. Compare the tilt with the same eligible stocks and base weights without the tilt.

Monthly, 10 bps	Net Sharpe	Max drawdown
Snapshot-cap base	1.119	-33.25%
Snapshot-cap + overlay	1.115	-32.51%
Equal-weight base	0.738	-38.34%
Equal-weight + overlay	0.728	-38.18%

2,390 return dates: 2015-07-06 to 2024-12-31. Monthly rebalance; 10 bps per dollar bought/sold; next-close execution. Sharpe uses a constant assumed daily risk-free return of 0.0001.



Primary snapshot-cap Sharpe difference: -0.0038; 95% paired temporal interval [-0.0289, +0.0194]. Mean block 21, 2,000 resamples. The small gross benefit disappears under the primary cost assumption.

What the implementation adds

Aligned holding periods, drifting weights, self-financing costs, trading-session rebalancing and explicit invalid-forecast fallback. An independent dollar-position calculation reconciles the primary portfolio results.

What the results do not establish

The later constituent list and repeated later cap snapshot create survivorship and look-ahead bias. Equal weighting does not fix that universe. Cached stock forecasts were not retrained. These are retrospective accounting diagnostics, not validated alpha; Part A's MSE change cannot explain Part B's Sharpe.

Full research report (8 pages) | Code and evidence

Sources: reports/additive_results.json and reports/portfolio_overlay_results.json; full raw data are not bundled. Based on collaborative research; brief prepared by Eddie Liu.